

 सत्यमेव जयते	भारत सरकार दूरसंचार विभाग दूरसंचार अभियांत्रिकी केंद्र Government of India Department of Telecommunications Telecommunication Engineering Centre K.L. Bhawan, Janpath, New Delhi- 110001	 ISO 9001:2015
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No. 26-1/2022-FA/TEC

Date: 26.10.2023

Subject: Minutes of Eighth (8th) meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group (SG)-15 held on 19.10.2023.

The eighth meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group 15 regarding “*Networks, Technologies and Infrastructures for Transport, access & home*”, reconstituted for the Study Period 2022-24, was held on 19.10.2023 under the chairpersonship of Sh. Rakesh Desai, DDG(FA), TEC through video-conferencing. List of participants is placed at Annexure-A.

2. At the outset, Sh. Deepak Pathak, Director (FA-I) TEC & Convener of NWG-15, welcomed all the participants and briefly explained about the agenda and further proceedings of the meeting.

3. Sh. Rakesh Desai, Chairman NWG-15 & DDG (FA) TEC, greeted all the members and welcomed all participants especially new members Mr. Shriram from M/s Aprecomm and Prof. Anand Srivastava from IIIT Delhi. Sh. Desai, in his opening remarks, conveyed that this meeting was convened to deliberate on the contributions on ongoing/new work items and to build consensus for their submission to forthcoming meeting of ITU-T Study Group 15 (SG15) scheduled at Geneva from 20 Nov – 1st December 2023.

After the introduction, as decided in the previous meeting, Sh. Desai requested the members to present their contributions for further deliberations and also requested other members to actively participate in the discussions on various contributions.

4. Contribution on- ‘ New work item for digitalization of operation and maintenance of optical fibre networks using distributed fibre optic sensing (DFOS)’ (Q5, 6, 7/15)

Contributors: Experts from M/s Sterlite Technologies Limited (STL) viz. S/Sh. Badri Gomatam; Nagaraju Bezawada; Abhishek Sati; Ashit Padhi; Sudipta Bhaumik and Sh.Rakesh Desai from TEC

Dr. Nagaraju from M/s STL, during his presentation, explained that ‘Distributed acoustic sensing (DAS)’ technology is specifically designed for measuring acoustic disturbances along the length of a fibre optic cable and it converts an optical fibre into thousands of extremely sensitive acoustic/vibration sensors. He further highlighted that DAS technologies find application in various industries and fields where continuous and accurate acoustic monitoring along extended lengths of optical fibre is valuable. He pointed out that DAS technology is also used in digitalizing operations and maintenance of optical fibre networks to detect events such as digging, excavation, construction activities etc.; for prevention of cable damage; network route tracing; prediction of network flaps in OSP layer-

etc. Mr Bhaumik also clarified that the subject work item is related to Q5,6 & 7/15 and hence, it is also proposed in the contribution to start with a Technical Report involving Q5, Q6 and Q7 experts.

After detailed deliberation, NWG-15 recommended to submit the contribution to the upcoming SG-15 meeting.

5. Contribution on 'Proposal for amendment in Rec.L.100 "Optical fibre cables for duct and tunnel application'. (Q5/15)

Contributors: Sh. Sudipta Bhaumik & Sh. Roshan Kumar from M/s Sterlite Technologies Limited and Sh. Vijay Dixit & Ms. Mudita Chandra from TEC

Sh. Bhaumik from M/s STL, during his presentation, explained that this contribution proposes amendment of Rec. L.100 in view of the TEC standards and also proposes to include new Annex B / Appendix I to the Rec. L.100 that describes some of the commonly used optical fibre cable designs for underground duct applications. The contributors were requested to furnish the updated draft by 25th October, 2023.

After detailed deliberation, NWG-15 recommended to update the draft to incorporate minor editorial corrections as discussed and submit the updated contribution to the upcoming SG-15 meeting.

6. Correspondence discussion for revision of Recommendation L.360: "Operations support system requirements for infrastructure and network elements management using Identification data (ID) technology" (Q7/15)

Contributors: Sh.Rakesh Desai, TEC and S/Sh. Guharajan Shivakumar & Sriram from M/s Aprecomm.

Sh.Desai apprised the participants that the Rapporteur of Q7/15 has initiated the correspondence discussions for the revision of Recommendation L.360: "*Operations support system requirements for infrastructure and network elements management using ID technology*". The Rapporteur (Q7/15) has also circulated a working document (WD) to Q7/15 experts seeking their inputs / comments wherein he added a new appendix II "Advanced technologies used for network infrastructure management" showcasing two technologies namely Image recognition technology and Network visualization technology. Sh. Desai mentioned that Virtual Network Expert (VNE) technology deployed in the FTTX networks of Indian ISPs is similar to the Network visualization technology and M/s Aprecomm has developed one of their software products on VNE technology. Draft inputs from NWG-15 in this regard proposing adding of Section II.3 to Appendix II explaining Virtual Network Expert (VNE) technology and its benefits have been discussed in the 7th meeting of NWG-15 and the updated draft was again circulated among NWG-15 members prior to the present meeting for further deliberations by the members in this meeting. Sh. Sriram presented the VNE technology to the participants and answered their queries. Sh.Desai pointed out that the Rapporteur (Q7/15) is preparing a TD (Temporary document) compiling all inputs received during correspondence discussion and this TD will be further discussed in the forthcoming meeting of SG15 scheduled from 20th Nov to 1st December 2023.

After detailed deliberation, NWG-15 recommended to communicate the Indian inputs to the Rapporteur (Q7/15) and Q7/15 experts in response to the ongoing correspondence discussion proposing to include Addendum for Virtual Network Expert (VNE) to

Appendix II of SG15-WD/WP2 on the Draft revised Recommendation L.360.

7. Questionnaire on Outdoor Optical Infrastructures sharing between multiple operators (Q7/15):-

Contributors: Sh. Rakesh Desai, TEC and Ms. Mudita Chandra, TEC

Sh. Rakesh Desai informed that the draft responses to the subject Questionnaire have been discussed in the last (7th) meeting of NWG-15 and as per the discussions in the meeting, responses of the NWG-15 have been uploaded on the designated portal of Q7/15 on 29/09/2023 prior to the prescribed deadline. He presented the participants the final responses to the said Questionnaire and also demonstrated the analysis prepared by the Rapporteur (Q7/15). He pointed out that the Rapporteur (Q7/15) is preparing a TD (Temporary document) compiling all responses and this TD will be further discussed in the forthcoming meeting of SG15 scheduled from 20th Nov to 1st December 2023.

8. Formation of NWP for NWG-15.

Chairman NWG-15 highlighted that the draft constitution of National Working party had been circulated with Minutes of 7th NWG Meeting wherein proposed members and IIT, Madras were requested to confirm their willingness for the proposed NWP-I & II. As responses from some of the members are still awaited, he stated that this matter will be decided in the next meeting.

9. Contributions for the forthcoming WTSA-24:-

Sh. Desai, while referring to earlier discussions, briefly explained the latest WTSA (World Telecommunication Standardization Assembly) resolutions and requested the members to examine the resolutions and propose amendments or new resolutions pertaining to SG15. He pointed out that next WTSA meeting is being held in Oct 2024 at New Delhi. He informed that first meeting of APT (Asia-Pacific Telecommunity) preparatory group for ITU WTSA-24 was held on 21st April 2023 and details are available at <https://www.apr.int/APT-WTSA24>. He further highlighted that in the upcoming meeting of SG15, there will be a special session regarding preparation of WTSA-2024. All NWG members were again requested to examine the current profile/responsibilities of SG15 and terms of references of the various Questions and submit inputs/ suggestions on WTSA resolutions related to SG15 and also related to general aspects like policy, procedures, industry participation, etc, by **15th November 2023** in view of the SG15 forthcoming meeting and second APT meeting scheduled in Jan 2024.

10. Use of Standards Coordination Portal

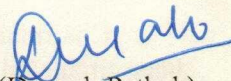
Sh. Rakesh Desai informed that significant numbers of NWG-15 members have already registered on 'Standards Coordination Portal' of TEC and further requested other members to register on 'Standards Coordination Portal' of TEC (<https://tec.gov.in/scp/>) and start familiarizing with the features of the portal. He further mentioned that important circulars, meeting notices and minutes of meeting will be posted on this portal only in due course.

11. Sh. Rakesh Desai also informed that Indian Mobile congress is scheduled on 27, 28 & 29th Oct 2023 at New Delhi and requested all members to visit Indian Mobile Congress 2023 to explore possible contributions for ITU-T SG15. He also requested all the NWG-15 members to register on ITU portal for ITU-T SG 15 meeting as last date of registration is 20th October, 2023. Sh. Desai also urged all NWG-15 experts to participate actively in group activities and submit more contributions.

12. The key action points emerged from the meeting are as under: -

- i. The finalised draft contributions will be uploaded on ITU-T website within the prescribed time on approval of the competent authority in D/o Telecommunications.
- ii. Members may examine the WTSA resolutions and propose amendments or new resolutions pertaining to SG15.
- iii. NWG-15 Members may please register on 'Standards Coordination Portal' of TEC (<https://tec.gov.in/scp/>) and start familiarizing with the features of the portal.
- iv. Members may study all the questions of ITU-T SG15 and come forward to increase number of Indian contributions in ITU-T SG15.

13. The meeting ended with the vote of thanks to the chair.


(Deepak Pathak)

Convener, NWG-15 & Director (FA-I), TEC

To,

All the participants / Members of the NWG-15.

Copy to: The Advisor / Sr.DDG, TEC for kind information.

List of Participants

1. Sh. Rakesh Desai, DDG (FA) & Chairman/NWG-15, TEC
2. Sh. Sudipta Bhuamik, M/s STL, Mumbai
3. Sh. N Murali Krishna, DDG (SR), TEC.
4. Sh. Shiv Narain, DDG, TEC
5. Dr. Nagaraju Bezawada, M/s STL
6. Sh. Srirama Murty Nistala, M/s Aprecomm
7. Prof Anand Srivastava, IITD, Delhi.
8. Sh. Rajeev Chaturvedi, Dura-Line India Ltd.
9. Sh. Roshan Kumar, M/s STL
10. Sh, Mandeep
11. Sh. Rana Pratap
12. Sh. R K Choudhary DDG, MH LSA, DoT, Pune
13. Sh. Venkata Rama Raju Chelle, Director, TEC.
14. Ms. Sonali, HFCL
15. Sh. Vivek Bohara
16. Sh. Deepak Pathak, Director (FA-I) TEC and convener NWG-15
17. Sh. Mohit Aggarwal, ADG, TEC.
18. Sh. Rakesh Goyal, ADG (QT), TEC
19. Sh. Ragunathan J, AD (FA), TEC.
